

Sanjivani Institutional LLM

A Domain-Specific Large Language Model for Higher Education

1. Introduction

Artificial Intelligence is becoming an important part of education, research, and institutional management. General-purpose Large Language Models (LLMs) such as ChatGPT and Gemini provide excellent general knowledge, but they do not possess detailed and updated information about a specific university. They cannot accurately answer institution-specific questions related to academics, university policies, examination rules, departments, notices, or administrative procedures.

To address this gap, we propose to develop Sanjivani Institutional LLM, a domain-specific Large Language Model built specifically for the Higher Education ecosystem with specialization in Sanjivani University. The model will be developed without using commercial AI APIs and will focus on learning from publicly available educational resources and institution-specific knowledge.

The proposed system aims to become an intelligent assistant for students, faculty members, researchers, and administrative staff by providing accurate responses related to university operations and academic activities.

2. Problem Statement

Students and faculty members frequently search for information related to academic regulations, university policies, examination rules, admissions, research guidelines, and institutional procedures. This information is usually distributed across multiple websites, PDF documents, notices, and circulars, making it difficult to access quickly.

Existing general-purpose LLMs are not trained specifically on Sanjivani University's knowledge and therefore cannot provide reliable institution-specific answers.

There is a need for an intelligent university-specific language model that understands the higher education domain and provides accurate information using trusted institutional resources.

3. Project Objective

The objective of this project is to design and develop a domain-specific Large Language Model for the Higher Education ecosystem, with specialization in Sanjivani University.

The proposed model will:

- Learn the language and terminology of higher education.
- Understand academic, administrative, and research-related information within a university.
- Be developed without relying on commercial AI APIs.
- Use publicly available educational resources along with institution-specific documents.
- Support Retrieval-Augmented Generation (RAG) for accessing updated university information.
- Provide an intelligent assistant for students, faculty, and administrative staff.

4. Proposed Domain

Instead of focusing on a single narrow topic, the proposed domain will be:

Higher Education & Institutional Knowledge

This is one unified domain consisting of three closely related subdomains:

- Academic Information
- Administrative Information
- Research Information

These areas naturally belong to the higher education ecosystem and together represent the institutional knowledge of a university.

5. Proposed Solution

The project proposes the development of a small-scale Large Language Model trained from scratch using educational text collected from publicly available sources.

The project will be completed in three phases.

Phase 1 – Base Model Development

Develop a transformer-based language model from scratch using publicly available higher education resources.

Public data sources may include:

- AICTE Guidelines
- UGC Regulations
- National Education Policy (NEP)
- Open university regulations
- Academic policies
- Educational documentation
- Public student handbooks

The purpose of this phase is to teach the model the language and terminology commonly used in higher education.

Phase 2 – Institutional Specialization

After developing the base model, institution-specific knowledge will be incorporated through continued training using Sanjivani University's publicly available and approved resources such as:

- Academic Calendar
- Student Handbook
- Examination Rules
- Course Syllabus
- Faculty Information
- Department Information
- Notices
- Circulars
- Research Guidelines
- University Policies

This phase enables the model to understand Sanjivani-specific terminology and institutional information.

Phase 3 – Knowledge Retrieval using RAG

To provide updated responses without retraining the model repeatedly, a Retrieval-Augmented Generation (RAG) framework will be integrated.

The RAG system will retrieve relevant information from university documents including:

- PDF documents
- Notices
- Circulars

- Policies
- Research papers
- Academic documents

The retrieved information will be used to generate accurate responses based on the latest available data.

6. Target Users

The proposed system is intended for:

- Students
- Faculty Members
- Administrative Staff
- Research Scholars
- Prospective Students

7. Expected Features

The proposed system will support:

- University Question Answering
- Academic Information Assistance
- Examination Guidance
- Administrative Information Support
- Research Information Assistance
- Document-based Question Answering
- Institution-specific Knowledge Retrieval
- Natural Language Interaction

8. Proposed Dataset Collection Strategy

The dataset will be collected in two stages.

Stage 1 – Public Dataset

Educational content available in the public domain.

Examples include:

- AICTE
- UGC
- NEP
- Open Educational Resources
- Public University Regulations
- Educational Policies

This data will help the model understand the Higher Education domain.

Stage 2 – Institutional Dataset

Institution-specific information collected with permission or from publicly available university resources.

Examples include:

- Student Handbook
- Academic Calendar
- Course Syllabus
- Faculty Directory
- Department Information
- Examination Rules
- Notices
- Circulars
- Research Information

This dataset will specialize the model for Sanjivani University.

9. Proposed Technology Stack

1. **Programming Language:** Python
2. **Deep Learning Framework:** PyTorch
3. **Model Architecture:** Transformer Decoder
4. **Tokenizer:** Custom Tokenizer
5. **Dataset Processing:** Custom Data Pipeline
6. **Knowledge Retrieval:** RAG Framework
7. **Vector Database:** FAISS / ChromaDB (Offline)
8. **Frontend:** Web-based Chat Interface
9. **Deployment:** Local Server
10. **External APIs:** None

10. Expected Deliverables

The project aims to deliver:

- Domain-Specific Large Language Model
- Higher Education Dataset
- Institution-specific Dataset
- Retrieval-Augmented Generation Framework
- Functional Prototype
- Technical Documentation
- Live Demonstration

11. Expected Outcomes

The proposed system is expected to:

- Provide accurate institution-specific responses.
- Reduce the effort required to search university documents.
- Improve access to academic and administrative information.

- Demonstrate practical implementation of a domain-specific Large Language Model.
- Create a foundation for future AI-based services within Sanjivani University.

12. Future Scope

The proposed model can be extended in future to support:

- Voice-based interaction
- Mobile application
- ERP integration
- Learning Management System (LMS) integration
- Student Portal integration
- Department-specific assistants
- Alumni and Placement support

Conclusion

The proposed Sanjivani Institutional LLM aims to develop a practical and domain-specific language model focused on the Higher Education ecosystem with specialization in Sanjivani University. By combining publicly available educational resources with institution-specific knowledge and an offline Retrieval-Augmented Generation framework, the project seeks to create an intelligent assistant that can support students, faculty, researchers, and administrative staff. This approach aligns with the competition objective of building a unique, functional, and educationally valuable LLM while providing a scalable foundation for future AI initiatives within the university.